



Atrium Healthcare Cost Transparency Database Development



Academic Integrity statement signed by all team members:

"In preparing this assignment, our team has observed the requirements of the UNCC Code of Student Academic Integrity. The submitted deliverables are entirely the work of our team."

Names	Roles & Responsibilities	Sign here to show that you have read and have complied to the UNCC Academic Integrity
Chris Taylor	Graphic Designer- Worked on creation of ERD including tables and attributes.	Christopher Taylor
Jackson Roberts	Database Administrator- created data dictionary as well as assisted in creation of the the ERD	Jackson Roberts
Sedi Kartey	Database Designer- Proofreader, editor, and assisted with ERD tables and relationships	Sedi Kartey
Truitt Evelyn	Contributed to the design and implementation of ERD, as well as business rules	Truitt Evelyn

Relevant files:

File Name	Description
Normalized Crows Foot ERD	Click Here for Full PDF
[file name]	
Optional	

Version control:

Version	Changes

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Part I. Database Initial Study

About the Model Company

Atrium Health is a not-for-profit healthcare system based in Charlotte, North Carolina, with hospitals, clinics, and care centers spread across the Southeast. It offers everything from routine checkups to advanced specialty care, while also supporting medical research and community health programs. Known for its innovation and wide network of providers, Atrium Health works to make quality healthcare more accessible for the communities it serves.

Problems and Constraints

Atrium Health currently faces challenges in its billing process, primarily due to a lack of upfront transparency that leaves patients unaware of treatment costs until after care. Insurance complexity adds to the problem, as co-pays, deductibles, and coverage limits are difficult for patients to understand in advance, often resulting in unexpected charges and dissatisfaction. Cost estimates also vary across departments, creating inconsistencies and eroding trust. The project is constrained by the need to comply with HIPAA regulations, integrate with existing hospital systems, and ensure accurate and timely cost data. Insurance variability further complicates billing, as different providers follow different rules. Additionally, scalability is critical for a large system like Atrium, and successful implementation will require staff training and overcoming resistance to change within a limited time and budget resources.

Project Objectives and Business Needs

This project aims to design and implement a comprehensive relational database system for Atrium Health to significantly enhance cost transparency for medical services by providing patients with clear, pre-service estimates of potential expenses, which addresses the critical business need to eliminate unexpected medical bills and empower patients to make informed healthcare decisions. The new system will also consolidate billing information, efficiently manage appointments, claims, and payments, and enable robust reporting capabilities to improve patient satisfaction, reduce administrative burdens, and ensure Atrium Health remains competitive and compliant in an evolving healthcare landscape.

Scope and Boundaries

Scope

The project will focus on improving Atrium Health's billing process by creating a database in MS Access that provides upfront cost estimates for patients. The system will store standardized treatment, procedure, and medication costs, calculate patient responsibility after insurance coverage, and generate clear billing records. It will also track payments, outstanding balances, and insurance claims to ensure both the patients and hospital staff have accurate and accessible financial information.

Boundaries

The database will not replace Atrium's existing electronic health record or scheduling systems but will function alongside them to support billing transparency. It will not handle complex medical data beyond what is necessary for cost calculation, nor will it serve as a full hospital management system. Insurance rules will be simplified for the project scope, and the system will be designed for demonstration purposes rather than enterprise-scale deployment.

Part II. Database Design

Business Rules

- **Patient ↔ EmergencyContact:** One patient can have many emergency contacts; each emergency contact belongs to exactly one patient.
- **Patient ↔ PreExistingCondition:** One patient can have many pre-existing conditions; each pre-existing condition belongs to exactly one patient.
- **Patient ↔ Treatment:** One patient can have many treatments; each treatment must belong to exactly one patient.
- **Patient ↔ Patient_InsurancePolicy:** One patient can hold many insurance policy records; each policy record belongs to exactly one patient.
- **InsurancePolicy ↔ Patient_InsurancePolicy:** One insurance policy can cover many patient records; each patient record references exactly one insurance policy.
- **Patient ↔ BillingEstimate:** One patient can have many billing estimates; each billing estimate belongs to exactly one patient.
- **Patient ↔ Invoice:** One patient can have many invoices; each invoice belongs to exactly one patient.
- **Patient ↔ Claim:** One patient can file many claims; each claim must belong to exactly one patient.

- **Patient ↔ Medication/Prescription:** One patient can have many prescriptions; each prescription belongs to exactly one patient.
-

- **Physician ↔ Department:** One department can have many physicians; each physician belongs to exactly one department.
 - **Physician ↔ Treatment:** One physician can provide many treatments; each treatment is performed by exactly one physician.
 - **Physician ↔ Medication/Prescription:** One physician can issue many prescriptions; each prescription is written by exactly one physician.
 - **Physician ↔ BillingEstimate:** One physician can generate many billing estimates; each billing estimate is tied to exactly one physician.
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- **Department ↔ Facility/Location:** One facility can have many departments; each department belongs to exactly one facility.
 - **Department ↔ Service/Procedure:** One department can offer many services; each service is linked to exactly one department.
-

- **Facility/Location ↔ Treatment:** One facility can host many treatments; each treatment occurs in exactly one facility.
-

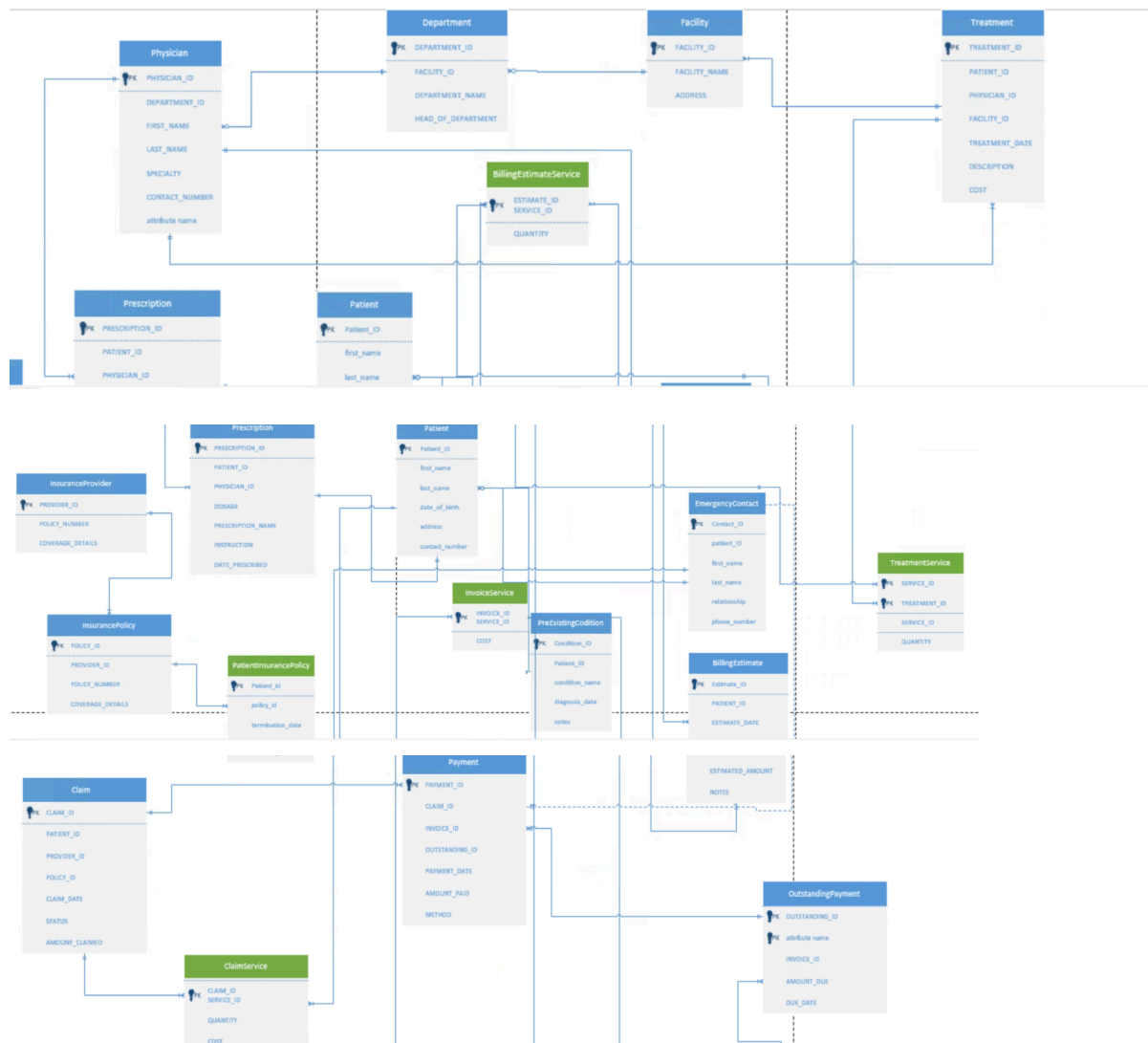
- **Service ↔ Treatment_Service:** One service can be part of many treatment records; each treatment record specifies exactly one service.
 - **Treatment ↔ Treatment_Service:** One treatment can consist of many services; each service line belongs to exactly one treatment.
 - **Service ↔ BillingEstimate_Service:** One service can appear on many billing estimate records; each billing estimate record specifies exactly one service.
 - **BillingEstimate ↔ BillingEstimate_Service:** One billing estimate can include many service items; each service item belongs to exactly one billing estimate.
 - **Service ↔ Invoice_Service:** One service can appear on many invoice records; each invoice record specifies exactly one service.
 - **Invoice ↔ Invoice_Service:** One invoice can include many service items; each service item belongs to exactly one invoice.
 - **Service ↔ Claim_Service:** One service can appear on many claim records; each claim record specifies exactly one service.
 - **Claim ↔ Claim_Service:** One claim can include many service items; each service item belongs to exactly one claim.
-

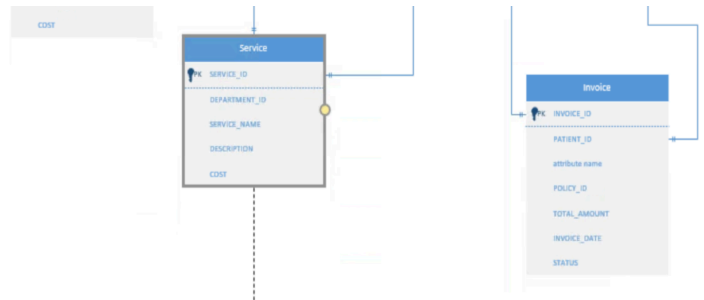
- **InsuranceProvider ↔ InsurancePolicy:** One insurance provider can offer many policies; each policy is offered by exactly one provider.
 - **InsuranceProvider ↔ Claim:** One insurance provider can receive many claims; each claim is submitted to exactly one provider.
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- **InsurancePolicy** ↔ **Claim**: One policy can be linked to many claims; each claim must reference exactly one policy.
- **InsurancePolicy** ↔ **Invoice**: One policy can be linked to many invoices; each invoice can be associated with exactly one policy.

- **Claim** ↔ **Payment**: One claim can generate many payments; each payment can be linked to exactly one claim.
- **Invoice** ↔ **Payment**: One invoice can have many payments; each payment is applied to exactly one invoice.
- **Invoice** ↔ **OutstandingPayment**: One invoice can have many outstanding payment records; each outstanding payment belongs to exactly one invoice.
- **OutstandingPayment** ↔ **Payment**: One outstanding payment can be reduced by many payments; each payment reduces exactly one outstanding payment.

Normalized ERD





Simplified Data Dictionary

Entity/Table Name	Attribute/Column Name	Key Type	Data Type	Normal Form	Justification / Description
Patient	patient_id	PK	INT	3NF	Unique identifier for the patient.
	first_name		VARCHAR		Patient's first name.
	last_name		VARCHAR		Patient's last name.
	date_of_birth		DATE		Patient's date of birth.
	address		VARCHAR		Patient's street address.
	contact_number		VARCHAR		Patient's phone number.
---	---	---	---	---	---
EmergencyContact	contact_id	PK	INT	3NF	Unique identifier for the emergency contact.
	patient_id	FK	INT		References Patient. One patient can have many contacts.

	<i>first_name</i>		VARC HAR		<i>Contact's first name.</i>
	<i>last_name</i>		VARC HAR		<i>Contact's last name.</i>
	<i>relationship</i>		VARC HAR		<i>Relationship to the patient.</i>
	<i>phone_number</i>		VARC HAR		<i>Contact's phone number.</i>
---	---	---	---	---	---
PreExistingCon dition	<i>condition_id</i>	PK	INT	3NF	<i>Unique identifier for the condition record.</i>
	<i>patient_id</i>	FK	INT		<i>References Patient.</i>
	<i>condition_name</i>		VARC HAR		<i>Name of the condition.</i>
	<i>diagnosis_date</i>		DATE		<i>Date the condition was diagnosed.</i>
	<i>notes</i>		VARC HAR		<i>Additional details.</i>
---	---	---	---	---	---
Physician	<i>physician_id</i>	PK	INT	3NF	<i>Unique identifier for the physician.</i>
	<i>department_id</i>	FK	INT		<i>References Department.</i>
	<i>first_name</i>		VARC HAR		<i>Physician's first name.</i>
	<i>last_name</i>		VARC HAR		<i>Physician's last name.</i>
	<i>specialty</i>		VARC HAR		<i>Physician's area of expertise.</i>
	<i>contact_number</i>		VARC HAR		<i>Physician's phone number.</i>
---	---	---	---	---	---
Department	<i>department_id</i>	PK	INT	3NF	<i>Unique identifier for the department.</i>

	facility_id	FK	INT		References Facility/Location.
	department_name		VARCHAR		Name of the hospital department.
	head_of_department		VARCHAR		Name of the person leading the department.
---	---	---	---	---	---
Facility/Location	facility_id	PK	INT	3NF	Unique identifier for the facility.
	facility_name		VARCHAR		Name of the facility.
	address		VARCHAR		Facility's street address.
---	---	---	---	---	---
Service/Procedure	service_id	PK	INT	3NF	Unique identifier for the medical service.
	department_id	FK	INT		References Department.
	service_name		VARCHAR		Name of the service.
	description		VARCHAR		Detailed description of the service.
	cost		DECIMAL		Standard cost of the service/procedure.
---	---	---	---	---	---
InsuranceProvider	provider_id	PK	INT	3NF	Unique identifier for the insurance provider company.
	provider_name		VARCHAR		Name of the insurance company.
	contact_info		VARCHAR		Contact details for the provider.
---	---	---	---	---	---
InsurancePolicy	policy_id	PK	INT	3NF	Unique identifier for the insurance policy.

	provider_id	FK	INT		References InsuranceProvider.
	policy_number		VARC HAR		Unique policy number.
	coverage_details		VARC HAR		Summary of policy coverage.
---	---	---	---	---	---
OutstandingPayment	outstanding_id	PK	INT	3NF	Unique identifier for the outstanding payment record.
	invoice_id	FK	INT		References Invoice.
	amount_due		DECIMAL		The remaining amount that is due.
	due_date		DATE		The date the outstanding amount is due.
---	---	---	---	---	---
Treatment	treatment_id	PK	INT	3NF	Unique identifier for the treatment episode.
	patient_id	FK	INT		References Patient.
	physician_id	FK	INT		References Physician.
	facility_id	FK	INT		References Facility/Location.
	treatment_date		DATE		Date the treatment occurred.
	description		VARC HAR		Details of the treatment provided.
	cost	Derived	DECIMAL		Derived Attribute: Total cost of treatment. Needed for transaction finality/reporting; usually stored despite being derivable from Treatment_Service.
---	---	---	---	---	---
Medication/Prescription	prescription_id	PK	INT	3NF	Unique identifier for the prescription.
	patient_id	FK	INT		References Patient.

	physician_id	FK	INT		References Physician.
	<i>medication_name</i>		VARC HAR		Name of the prescribed medication.
	<i>dosage</i>		VARC HAR		Amount and frequency.
	<i>instructions</i>		VARC HAR		Usage instructions.
	<i>date_prescribed</i>		DATE		Date the prescription was written.
---	---	---	---	---	---
BillingEstimate	estimate_id	PK	INT	3NF	Unique identifier for the billing estimate.
	patient_id	FK	INT		References Patient.
	physician_id	FK	INT		References Physician.
	<i>estimate_date</i>		DATE		Date the estimate was generated.
	estimated_amount	Derived	DECI MAL		Derived Attribute: Estimated total cost. Calculated from BillingEstimate_Service.
	<i>notes</i>		VARC HAR		Notes on the estimate.
---	---	---	---	---	---
Invoice	invoice_id	PK	INT	3NF	Unique identifier for the invoice.
	patient_id	FK	INT		References Patient.
	policy_id	FK	INT		References InsurancePolicy.
	<i>invoice_date</i>		DATE		Date the invoice was issued.
	total_amount	Derived	DECI MAL		Derived Attribute: Grand total amount charged. Calculated by summing the cost in Invoice_Service. Needed for transactional integrity.
	<i>status</i>		VARC HAR		Current status (e.g., 'Pending', 'Paid').

---	---	---	---	---	---
Claim	claim_id	PK	INT	3NF	Unique identifier for the insurance claim.
	patient_id	FK	INT		References Patient.
	provider_id	FK	INT		References InsuranceProvider.
	policy_id	FK	INT		References InsurancePolicy.
	claim_date		DATE		Date the claim was filed.
	status		VARC HAR		Status of the claim.
	amount_claimed	Derived	DECI MAL		Derived Attribute: Total monetary amount claimed. Calculated by summing the cost in Claim_Service. Needed for immediate financial tracking.
---	---	---	---	---	---
Payment	payment_id	PK	INT	3NF	Unique identifier for the payment transaction.
	claim_id	FK	INT		References Claim (Nullable).
	invoice_id	FK	INT		References Invoice (Nullable).
	outstanding_id	FK	INT		References OutstandingPayment (Nullable).
	payment_date		DATE		Date the payment was received.
	amount_paid		DECI MAL		Amount of money paid.
	method		VARC HAR		Payment method.
---	---	---	---	---	---
Patient_InsurancePolicy	patient_id	FK, Comp PK	INT	3NF	References Patient. Part of Composite PK.

	policy_id	FK, Comp PK	INT		References InsurancePolicy. Part of Composite PK.
	effective_date		DATE		Date coverage began for the patient under this policy.
	termination_date		DATE		Date coverage ended for the patient under this policy.
---	---	---	---	---	---
Treatment_Service	treatment_id	FK, Comp PK	INT	3NF	References Treatment. Part of Composite PK.
	service_id	FK, Comp PK	INT		References Service/Procedure. Part of Composite PK.
	quantity		INT		Number of times the service was performed in the treatment.
---	---	---	---	---	---
BillingEstimate_Service	estimate_id	FK, Comp PK	INT	3NF	References BillingEstimate. Part of Composite PK.
	service_id	FK, Comp PK	INT		References Service/Procedure. Part of Composite PK.
	quantity		INT		Number of times the service is included in the estimate.
---	---	---	---	---	---
Invoice_Service	invoice_id	FK, Comp PK	INT	2NF	References Invoice. Part of Composite PK.
	service_id	FK, Comp PK	INT		References Service/Procedure. Part of Composite PK.
	quantity		INT		Number of times the service is listed on the invoice.

	cost		DECIMAL		Justification (Not 3NF): cost depends only on service_id (partial dependency, violation of 3NF if cost is supposed to be the standard service cost). However, in billing, the cost must be stored here because the unit price for a service may change over time or be discounted on a specific invoice, thus relying on the full composite key for historical accuracy. This makes it a practical, denormalized design for audit trails.
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Claim_Service	claim_id	FK, Comp PK	INT	2NF	References Claim. Part of Composite PK.
	service_id	FK, Comp PK	INT		References Service/Procedure. Part of Composite PK.
	quantity		INT		Number of times the service is included in the claim.
	cost		DECIMAL		Justification (Not 3NF): Similar to Invoice_Service, cost might violate 3NF if it's the standard price. Practically, the specific cost of the service item at the time of claiming must be stored here for claim processing and audit, making it a necessary deviation from pure 3NF to ensure transactional immutability.

Notes explaining design decisions and assumptions

- The database is designed mainly in 3rd Normal Form (3NF) to reduce redundancy.
- Some tables use controlled denormalization (like **Invoice_Service** and **Claim_Service**) to keep historical and financial accuracy.
- Primary keys (PK) uniquely identify each record; foreign keys (FK) link related entities.
- Derived fields (like total cost or amount) are stored only when needed for reporting or audit purposes.
 - All foreign keys maintain referential integrity.
 - Derived totals (like invoice amounts) are stored for audit consistency.
 - Bridge tables handle all many-to-many relationships.

Part III. Implementation and Loading

This part is done in MS Access. Please see the relevant file.

Appendices

Business Forms

Patient Registration Form	
Patient information	
Full name:	Date of birth:
Gender: ☐ Male ☐ Female ☐ Other:	
Marital status: ☐ Single ☐ Married ☐ Divorced ☐ Widowed	
Address:	
Phone number:	Email address:
Emergency contact	
Name:	
Relationship to patient:	Phone number:
Insurance information	
Primary insurance:	
Policyholder's name:	
Policy number:	Group number:
Secondary insurance (if applicable):	
Policyholder's name:	
Policy number:	Group number:
Primary care physician information	
Physician name:	
Practice/clinic name:	Phone number:
Reason for visit	

Insurance Claim Form	
Please fill out the form to submit your insurance claim.	
Full Name	
First Name	Last Name
Policy Number	
Date of Incident	
Date	
Type of Claim	
Description of Incident	
Upload Supporting Documents	
Upload a File Drag and drop files here	
Contact Phone Number	
Please enter a valid phone number.	
Submit	

Above: Insurance Claim Form

Business Reports

[illegible]

Patient Visit Summary		Pebbles Flintstone (10 yrs, 1 mo; F; 04/03/04)		
Pebbles Flintstone DOB: 04/03/04 Sex: Female		PCC Pediatric Test Associates 20 Winoski Falls Way, Suite 7 Winoski, VT 05404 (800) 722-7708		PCC #: 3336
Scheduled Visits: None		Date of Last Physical: 04/11/13		
Visit Summary for 05/02/14				
Beverly Crusher MD. PCC Pediatrics Main Office Sick - Bright Futures				
Chief Complaint				
sore throat				
Vitals				
Temperature: 102 F				
 Labs				
Rapid Strep				Status: Completed
Facility: PCC Pediatric Test Associates (Doctor's Office)				
Test	Result	Units of Measure	Reference Range	Interpretation
Streptococcus pyogenes Ag In Throat by Immunoassay	Positive		Negative	Abnormal
Throat Culture				Status: Ordered
Result:				
Facility: PCC Pediatric Test Associates (Doctor's Office)				
Plan				
Medication				
Amox 10 days				
Other Health Information as of 05/05/14				
Active Problems		Onset Date		Problem Notes
Asthma		02/02/08		Sample problem note
Obesity		05/05/12		Sample problem note
Active Allergies		Onset Date		Reaction
Allergy Cat Hair				
PCC Pediatric Test Associates: 05/05/2014 11:56:07				Page 1 of 2

Above: Billing Summary Report

Above: Patient Visit Summary Report